

Endogenous Reachability Collapse (ERC-D)

A Minimal Dynamical Model of Loss of Recoverability Without Loss of State Space

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AI-assisted modeling and analysis using ChatGPT and Grok

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Abstract

We present a minimal dynamical model in which loss of recoverability may emerge without removal of the underlying state space. Under identical external input, different modules exhibit distinct recovery outcomes depending on internal parameters. A sharp transition in recoverability appears as a function of a sensitivity parameter, separating regimes where trajectories return to baseline from those where they remain in a high-deviation state. Notably, this transition occurs without elimination of the baseline state, consistent with the possibility that functional failure may arise from loss of dynamic accessibility rather than structural loss. The boundary is robust to threshold definition, feedback strength, and input duration, and disappears in the absence of nonlinear feedback. The model is intentionally minimal and designed to be falsifiable.

1. Introduction

In many models of system failure, dysfunction is associated with the loss or degradation of system components or states. However, it remains possible that failure may, in some cases, arise from a loss of access to otherwise preserved states.

Here we explore this possibility using a minimal dynamical system. The goal is not to reproduce a specific biological mechanism, but to demonstrate that loss of recoverability may emerge endogenously from system dynamics, even when the baseline state remains present.

2. Model

We consider a minimal dynamical system of the form:

$$\frac{dx}{dt} = -rx + s \cdot u(t) + f(x)$$

where:

- $x(t)$ represents deviation from baseline
- r is a recovery rate
- s is a sensitivity parameter
- $u(t)$ is a shared external perturbation
- $f(x)$ is a nonlinear feedback term introducing state-dependent dynamics capable of generating bistable-like behavior without explicitly removing any state

Multiple modules are defined with different parameter values but are subjected to the same external input.

3. Results

3.1 Differential recoverability under identical input

Under identical perturbation $u(t)$, different modules exhibit distinct recovery outcomes. Some trajectories return to baseline, while others remain in elevated states after the perturbation ends.

This separation is captured by metrics such as:

- mean deviation
- high-state occupancy
- late persistence
- recovered fraction

These metrics consistently distinguish modules that recover from those that do not.

3.2 Loss of recoverability without loss of state space

In all cases, the baseline state remains part of the system's state space. However, certain parameter regimes lead to trajectories that no longer re-enter this state.

This is consistent with failure occurring through loss of dynamic accessibility rather than structural removal of states.

3.3 Existence of a recoverability boundary

A parameter sweep over the sensitivity parameter s reveals a sharp transition in recoverability.

Below a critical value s^* trajectories return to baseline.
Above it, trajectories remain in a high-deviation regime.

In the present parameterization:

$$s^* \approx 0.13$$

This transition is abrupt, consistent with a boundary separating dynamically accessible and inaccessible regimes.

The boundary is robust to variations in the recovery threshold used to define return, persists across different values of the nonlinear feedback strength, and shifts with changes in input duration.

3.4 Role of nonlinear feedback

The recoverability boundary disappears when the nonlinear feedback term is removed ($b = 0$), resulting in uniform recovery across all values of the sensitivity parameter.

This suggests that the boundary does not arise from the external input alone, but from the internal nonlinear structure of the system.

3.5 Early-warning signals of reachability collapse

To further characterize proximity to collapse during system evolution, we define a dynamic safety margin:

$$S(t) = a(c(t)) - r(t)$$

where $r(t)$ represents the instantaneous trajectory radius and $a(c(t))$ the moving recoverability boundary.

Positive values indicate that the system remains within the recoverable region, values approaching zero indicate proximity to the regime boundary, and negative values indicate loss of dynamic accessibility to the baseline state.

In simulations, $S(t)$ decreases smoothly as internal constraint builds and crosses zero at the onset of collapse. The time derivative dS/dt provides an additional signal, showing a marked negative trend prior to loss of recoverability.

These observations suggest that reachability collapse may be preceded by continuous dynamical signatures, rather than occurring as an abrupt or purely stochastic transition.

4. Falsifiable Predictions

The model leads to the following predictions:

1. **Same input, different recovery**
Systems exposed to similar perturbations may exhibit different recovery outcomes depending on internal parameters.
2. **Threshold behavior**
Recoverability may change abruptly as a function of internal sensitivity or feedback strength.

3. **Metric separation**
High-state occupancy and late persistence may correlate with loss of recoverability.
4. **Dependence on internal dynamics**
If recoverability depends only on total input and remains insensitive to internal parameters or temporal structure, the model would not be supported.

Critical falsifier:

If recoverability varies smoothly and continuously with internal parameters, without any threshold-like transition, the proposed mechanism would be challenged.

5. Discussion

This minimal model is consistent with the possibility that system failure may arise not from disappearance of viable states, but from loss of dynamic access to them.

The disappearance of the boundary in the absence of nonlinear feedback suggests that loss of recoverability is not solely determined by external input, but may emerge from internal system structure.

While the model is abstract, it may serve as a conceptual framework for interpreting phenomena in which recovery becomes impaired without obvious structural damage.

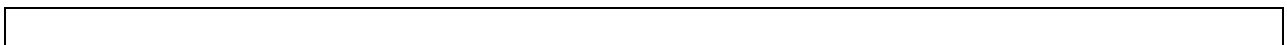
Further work is required to connect these dynamics to specific physical or biological systems.

6. Conclusion

We show that a minimal dynamical system can reproduce loss of recoverability without loss of state space. A sharp boundary separates regimes where return to baseline is possible from those where it is not. This is consistent with the possibility that functional failure may emerge from changes in dynamic accessibility rather than structural loss.

Figures

Figure 1
Differential dynamics under shared perturbation



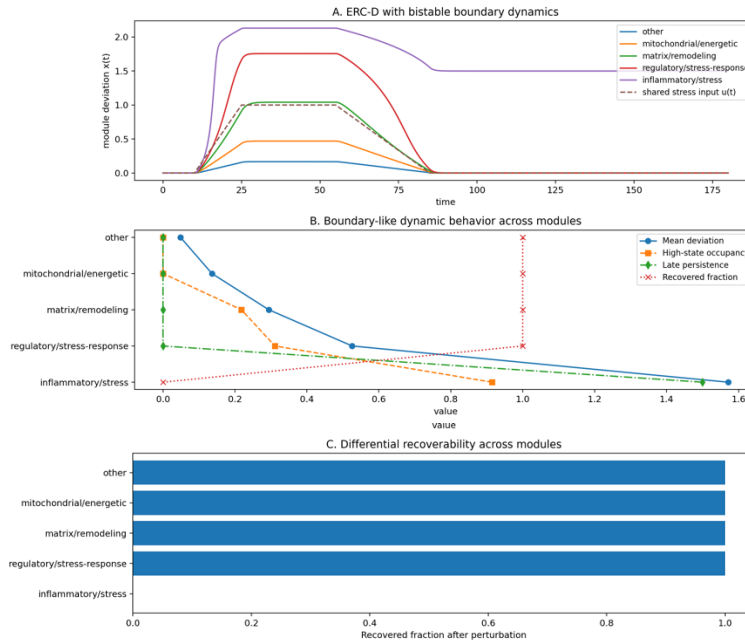


Figure 1. Differential dynamics under shared perturbation.

(A) Time evolution of system state $x(t)$ for multiple modules under identical external input $u(t)$. Despite identical forcing, modules exhibit distinct trajectories, with some returning to baseline and others remaining in elevated states.

(B) Summary metrics across modules, including mean deviation, high-state occupancy, late persistence, and recovered fraction. These metrics separate recovering from non-recovering regimes.

(C) Final recovered fraction across modules, illustrating differential recoverability under identical input conditions.

Together, these results show that recovery outcomes depend on internal parameters rather than external input alone.

Figure 2

Recoverability boundary as a function of sensitivity parameter ($s^* \approx 0.133$)

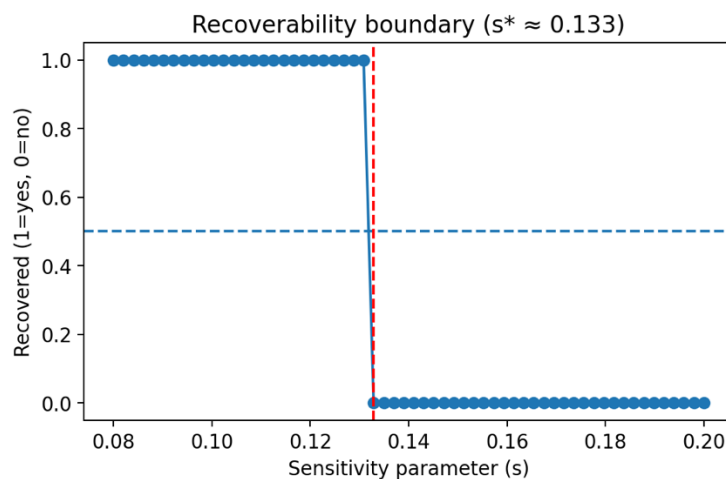


Figure 2. Recoverability boundary as a function of sensitivity parameter.

Recovered fraction is shown as a function of the sensitivity parameter s .

A sharp transition is observed at a critical value $s^ \approx 0.13$, separating regimes where trajectories return to baseline from those where recovery fails.*

The dashed horizontal line indicates the threshold used to define recovery, and the vertical dashed line marks the estimated boundary s^ .*

This abrupt transition is consistent with a boundary between dynamically accessible and inaccessible regimes, rather than gradual degradation of recovery.

The boundary disappears in the absence of nonlinear feedback ($b = 0$), confirming its endogenous origin.

Data & Code

This study uses synthetic data generated from a minimal dynamical model. No external or empirical datasets were used.

Code and data are available in the associated repository.